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Large eddy simulation of a model gas turbine combustor

F. di Mare,[a] W.P. Jones,[a,*] and K.R. Menzies[b]

[a] *Department of Mechanical Engineering, Imperial College London, Exhibition Road, London SW7 2AX, UK*
[b] *Combustion Systems Engineering, Rolls-Royce plc., PO Box 3, Filton, Bristol BS34 7QE, UK*

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Abstract

Large eddy simulation (LES) has been used to predict temperature and species concentrations in a model can-type gas turbine combustor operating in a non-premixed combustion regime. The subgrid scale stresses have been modelled adopting the standard Smagorinsky–Lilly model, whilst combustion has been accounted for using a conserved scalar approach. The dependence of the thermodynamical status of the mixture on the conserved scalar has been described invoking a flamelet assumption, and the shape of the scalar probability density function (pdf) assigned to be a β function. The subgrid scalar variance, needed to parameterise the scalar pdf, has been modelled according to local equilibrium arguments. The complex flow pattern developing from the interaction of a strongly swirling flow in the primary zone with the impinging primary jets has been captured by the present simulations in great detail. In particular, it has been observed that an accurate description of the inlet section of the combustor plays a fundamental role in the prediction of the fuel placement and, hence, of the temperature distribution in the primary region.
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Keywords: Combustion; Turbulence; Large eddy simulation; Gas turbines

1. Introduction and motivation

The majority of current methods for calculating the properties of turbulent flames and practical combustion systems are based on Reynolds or Favre averaging of the Navier–Stokes equations (RANS). Whilst the approach has often proved useful in the past it has met with variable success in reproducing many of the important effects present in practical combustion systems, particularly in situations where large-scale organised and nonstationary turbulence structures, strong streamline curvature, or nongradient transport is present. Large eddy simulation (LES) represents a potentially powerful and promising method for overcoming many of the limitations of RANS approaches. LES involves a direct three-dimensional time-dependent computation of the large-scale turbulent motions responsible for turbulent mixing whilst those with scales smaller than the computational grid (the subgrid stresses) are modelled. Recently, more emphasis has been placed on applications of LES to combusting flows, the major issue being the fact that reactive phenomena occur on scales that fall well below the resolution limit of the computational mesh and as a consequence modelling is required. This difficulty has been addressed by several workers with limited success, for example, see [1–4]; in these approaches direct closure of the chemical source term in the species transport equations is attempted. More sophisticated techniques have been also proposed, including, for example, the application of the linear eddy model [5] to LES by Menon et al. [6], the extension of the conditional moment closure, devised originally in the context of RANS [7], to LES by Bushe and Steiner [8], and the solution of modelled scalar pdf transport equations, introduced by

* Corresponding author. Fax: +44(0) 207 581 54 95.
E-mail address: w.jones@imperial.ac.uk (W.P. Jones).

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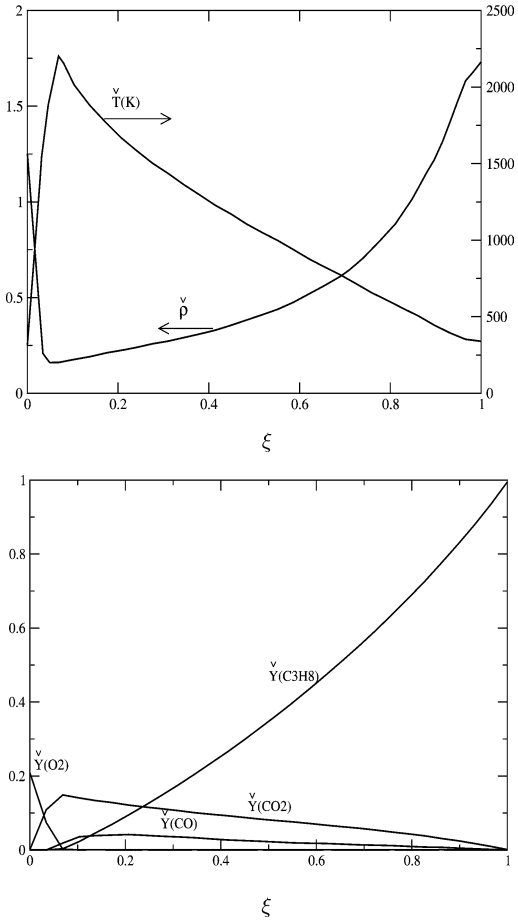

Fig. 1. Dependence of species mole fraction, temperature, and density on the mixture fraction: laminar flamelet calculation ($a_s = 15\ s^{-1}$).

Givi and co-workers [9,10] and then successively developed in [11–14]. The potentially greater degree of accuracy offered by these models is limited by the extremely high computational cost of implementation. So far this has restricted their use to very simple and idealised problems; they do not currently represent a solution to practical problems.

An alternative to the above is a simple laminar flamelet conserved scalar approach, which provides a reasonable compromise between implementation costs and accuracy (Fig. 1). Here it is presumed that a single strictly conserved scalar, the mixture fraction ξ, is sufficient to describe the thermochemical state of the flow and this serves to de facto decouple the modelling of turbulence from that of reactive phenomena. The methodology originally developed in the context of RANS approaches (e.g., [15]) has then been applied to LES in a slightly modified form (see [16–18]). The functional dependence of composition, temperature, and density on ξ can be obtained either from chemical equilibrium assumptions or from laminar flamelet calculations. Although the conditions for flamelet combustion may not always be satisfied, the method yields more realistic predictions of species concentrations (particularly in the case of CO) for hydrocarbon fuels than the equilibrium method and it is therefore to be preferred. The mean (filtered) concentrations, temperature, and density can be calculated by convolution of these known functions with the subgrid pdf of the resolved conserved scalar. The functional form of the probability density function can be conveniently chosen as a β function, which is defined in the interval [0, 1], as is the mixture fraction itself. Other pdf forms have also been used (see [19]) such as clipped gaussian and double delta functions. Although relying on assumptions that may not be always strictly valid, the conserved scalar approach is computationally inexpensive, particularly in the context of full three-dimensional time-dependent computations.

The application of LES to gas turbine combustors is being regarded with increased interest due to its potential in capturing features of the flow field that are currently out of reach of other modelling techniques. However, the complex geometrical aspects of common industrial devices, which drastically increases the cost of performing good-quality LES, together with the issues related to incorporating sufficiently accurate combustion models still pose a limitation on the application of LES to practical gas turbine combustion chambers. Several authors have attempted LESs of gas turbine combustors in simplified geometries. Kim et al. [20] calculated the velocity and composition field in a GE lean premixed dry-low NO_x LM6000 combustion chamber using a flamelet approach to describe the premixed combustion occurring in the combustor. Although the device retained some of the complex characteristics of practical gas turbine combustors, film cooling was not accounted for, nor were additional complications related to the presence of dilution jets or changes in the shape of the chamber considered. Also, the computed species concentration and temperature were not reported, presumably because no measurements were available for comparison purposes. Mahesh et al. [21] also carried out LES of a simplified Pratt & Whitney coaxial combustor, mainly for code validation purposes, and Angelberger et al. [22] investigated the effects of acoustic instabilities in a premixed dump combustor.

Although a premixed combustion regime is becoming increasingly popular for industrial gas turbines, because of their potential for very low NO_x emissions, combustion in conventional aircraft gas turbines occurs predominantly in the non-premixed diffusion flame regime. In such devices turbulent mixing exerts a dominant influence and better understand-

ing necessarily requires an accurate prediction of the turbulent velocity field, including any "unsteady" aspects associated, for example, with jet-on-jet impingement. The present study represents an attempt to predict the concentration, temperature, and velocity field in a practical configuration where the combustion process develops in the non-premixed regime. The configuration selected in the present work is a model can-type combustor closely representative of that found in the Rolls–Royce Tay engine. The combustor, described in [23–25], comprises rows of primary and dilution ports, a circular to rectangular nozzle, as well as of porous walls, and the resulting flow pattern is extremely complex and is well suited to an investigation of the capabilities of LES. In the present work, the conserved scalar approach has been adopted to model the combustion process, both in view of the advantages offered by the method over more expensive, although more rigorous alternatives, and because of its success in previous works [16,17,26]. The Smagorinsky–Lilly model [27,28] has been used to represent the subgrid stresses and fluxes and thus close the filtered momentum and conserved scalar transport equations. In the following sections the details of the mathematical model are described, along with the numerical method used to solve the governing equations, and results of the computation are then compared with measurements.

1.1. Flow configuration

The device here investigated is a roughly three-fourths-scale model can-type combustion chamber representative of the Rolls–Royce Tay gas turbine. The model combustor comprises a hemispherical head, a cylindrical barrel, and a circular to rectangular nozzle. The combustor walls are made of *Transply*, a laminated porous material engineered at Rolls–Royce plc. A relatively small swirler with 18 aerodynamically shaped vanes sits in the head of the combustor, and fuel (high-purity propane) is injected into the combustion chamber through ten 1.7-mm-diameter jets arranged around a 90° cone located at the centre of the swirler. Two rows of primary and dilution ports are located in the combustor barrel, with the latter staggered by half a pitch with respect to the former. The diameter of the dilution jets is twice that of the primary ones. The combustor is located in a large duct and is effectively "plenum fed" so that the primary and dilution jets enter the combustor at approximately 90° to its axis. The combustor geometry is described in greater detail in [23,29]. A schematic representation of the combustor is shown in Fig. 2.

In an isothermal experimental study conducted at Imperial College, two different swirler configurations were investigated, one featuring 30° vanes, the other

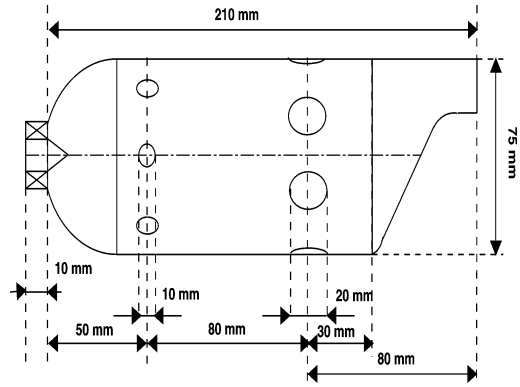

Fig. 2. Schematic representation of the model Tay combustor.

45° vanes. The influence of swirler characteristics on the flow field inside the combustor was assessed by Bicen and Palma [30]. They found that the swirler flow characteristics are unaffected by the Reynolds number in the range at which the combustor was operated. There was also evidence of a stronger stabilising effect of the near-solid body rotation to which the fluid is subjected in proximity of the combustor axis on turbulence when the 45° swirl was used. The lower discharge coefficient C_d found in this case seemed to suggest that separation occurred in the vane passage, which was also the cause of difference between the flow split with the two swirler configurations. Detailed combusting flow measurements within the combustor have been taken for a number of different simulated operating conditions, encompassing take off and ground-idle regimes, at nearly atmospheric pressure. Temperature was measured by digitally compensated small-diameter bare-wire thermocouples while concentrations of UHC, CO_2, CO, O_2, and H_2 were obtained by sampling through water-cooled probes and analysing the samples with a flame ionisation detector, a gas chromatograph, and infrared and paramagnetic analysers [23].

In the present study computations were carried out for the configuration featuring a 45° swirler operating under the takeoff conditions, for which the initial air temperature was 315 K and the air-to-fuel-ratio (AFR) 57. This corresponds to volumetric flow rates of 0.08 and 0.001 $m^3 s^{-1}$ for air and fuel, respectively, yielding a bulk velocity equal to 182.2 ms^{-1} for the former and 44.1 ms^{-1} for the latter, based on cold flow properties. The Reynolds number based on the characteristic dimensions of the swirler is 79,900 for air. The Reynolds number based on the bulk air flow rate and barrel diameter is 79,000. The mass flow rates through the Transply walls and dilution and primary ports were distributed so that 6.6% of the air intake went through the head of the combustor, 13.8%

through the barrel's wall, 5.8% through the nozzle, and 13.6 and 53.3% through the primary and dilution ports, respectively. A characteristic residence time for the combustor can also be defined as the ratio of the volume of the combustor to the bulk volumetric flow rate of air, resulting in $\tau_r = 0.01012$ s.

2. Mathematical model

In LES a spatial filter is applied to the equations of motion so that motions with scales less than the filter "width" do not have to be resolved by the computational mesh. The spatial filter of a function $f = f(\mathbf{x}, t)$ is defined as its convolution with a filter function, G, according to

$$\bar{f}(\mathbf{x}, t) = \int_{\Omega} G\big(\mathbf{x} - \mathbf{x}'; \Delta(\mathbf{x})\big) f(\mathbf{x}', t)\, d^3x', \tag{1}$$

where the integration is carried out over the entire flow domain, Ω. The filter function has a width Δ which may vary with position. While a number of different filter kernels have been proposed and used, e.g., "sharp spectral cutoff" and gaussian and box (or top-hat) filters, in combustion the filter function must satisfy

$$G(x - x', \Delta) > 0. \tag{2}$$

Also, in turbulent reacting flows large density variations occur that must be properly accounted for and this is achieved by the introduction of density-weighted filtered quantities, denoted by a tilde and defined by $\tilde{f}(\mathbf{x}, t) \equiv \overline{\rho f}/\bar{\rho}$.

Application of the density-weighted filter to the equations of motion gives

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial \bar{\rho}\tilde{u}_i}{\partial x_i} = 0, \tag{3}$$

$$\frac{\partial \bar{\rho}\tilde{u}_i}{\partial t} + \frac{\partial \bar{\rho}\tilde{u}_i\tilde{u}_j}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu\left[\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i}\right]\right) - \frac{\partial \tau_{ij}}{\partial x_j}, \tag{4}$$

where τ_{ij} is the unknown *residual* or *subgrid scale* stress, defined as

$$\tau_{ij} = \bar{\rho}(\widetilde{u_i u_j} - \tilde{u}_i\tilde{u}_j). \tag{5}$$

To evaluate the viscous terms in Eq. (4) subgrid fluctuations in the density and the viscosity μ are neglected. Direct viscous stresses are negligible in the high-Reynolds-number flow presently to be considered. To close Eq. (4) the standard Smagorinsky–Lilly model [27,28] has been adopted:

$$\tau_{ij} - \tfrac{1}{3}\tau_{kk}\delta_{ij} = 2\rho C_s \Delta^2 \|\tilde{S}_{ij}\|\tilde{S}_{ij}. \tag{6}$$

Here $\|\tilde{S}_{ij}\|$ represents the Frobenius norm of the Favre filtered rate of strain tensor, $\tilde{S}_{ij}$. The filter width Δ is equated here to the cube root of the local grid cell volume. The isotropic part of the viscous and subgrid stresses is adsorbed into the pressure. In writing Eq. (4) it is has been assumed that the filter width is invariant in space. However, this is not the case in practical applications, where the need for different levels of spatial resolution across the solution domain requires the use of variable grid spacings and this results in a *commutation error*. This commutation error can be shown to be almost entirely dissipative in nature [31], and in the present work estimates of this "dissipation" have been demonstrated [32] to be negligible compared with dissipation of the SGS model itself.

2.1. Combustion model

In principle, the system of governing equations could include the transport equations for chemical species, according to the assigned reaction mechanism, and energy. However, this is a difficult and computationally expensive task not only because of the dramatic increase in the number of equations to be solved, but also due to the presence of nonlinear filtered chemical source terms that must be modelled (e.g., [26]).

In situations such as the present, however, the fuel and air streams enter the combustion chamber in separate streams and heat-releasing reactions may be presumed "fast." With a number of fairly benign assumptions—the main ones being that the flow is adiabatic, the Prandtl and Schmidt numbers are equal, and the pressure is thermodynamically constant—the instantaneous major species composition, temperature, and density can be related to a strictly conserved scalar quantity, the mixture fraction ξ [15,33, 34]. The mixture fraction can be viewed as a normalised chemical element mass fraction or enthalpy defined so that it takes the value of unity in the fuel stream and zero in the air stream. The equation for ξ contains no reaction source terms, is affected only indirectly by reaction through its influence on fluid density, and is hence strictly conserved. In gas turbine combustors the Mach number is low, the pressure is approximately constant, and heat losses to the walls are negligible [35], and as a consequence the conserved scalar approach is appropriate. The conservation equation for the Favre filtered mixture fraction can then be written [17] as

$$\frac{\partial \bar{\rho}\tilde{\xi}}{\partial t} + \frac{\partial \bar{\rho}\tilde{u}_i\tilde{\xi}}{\partial x_i} = \frac{\partial}{\partial x_i}\left(\frac{\mu}{\sigma}\frac{\partial \tilde{\xi}}{\partial x_i}\right) - \frac{\partial \bar{\rho}(\widetilde{u_i\xi} - \tilde{u}_i\tilde{\xi})}{\partial x_i}, \tag{7}$$

where σ is the Prandtl/Schmidt number. Eq. (7) contains the unknown subgrid scalar flux,

$$J_i^{\mathrm{sgs}} = \bar{\rho}(\widetilde{u_i\xi} - \tilde{u}_i\tilde{\xi}), \tag{8}$$

which is modelled using a gradient transport model:

$$J_i^{\mathrm{sgs}} = -\frac{\mu_{\mathrm{sgs}}}{\sigma_{\mathrm{sgs}}}\frac{\partial\tilde{\xi}}{\partial x_i}. \tag{9}$$

There is some uncertainty in the choice of the value of the subgrid Prandtl/Schmidt number. If a convective-inertial (Obukhov–Corrsin) spectrum is assumed for the SGS scalar variance $\tilde{\xi}''^2_{\mathrm{sgs}}$, defined as $\tilde{\xi^2} - \tilde{\xi}^2$, the value for the subgrid Prandtl/Schmidt number can be determined by invoking an equilibrium assumption. On this basis Schmidt and Schumann [36] proposed a value $\sigma_{\mathrm{sgs}} = 0.42$, whereas other authors ([37] for instance) used a value of 0.36. In view of this uncertainty some workers [38,39] have used a dynamic calibration procedure for σ_{sgs} similar to that devised by Germano et al. [40] for the Smagorinsky model parameter. In the present case, however, based on the results of [16,17,41], a fixed value equal to 0.7 has been adopted.

The functional dependence of the thermochemical variables on the mixture fraction has been obtained from a flamelet assumption. In "steady" flamelet approaches it has been usual in the past for this functional dependence to be made dependent also on some measure of flame stretch, usually taken to be the "scalar" dissipation rate [42,43]. However, even in RANS-based methods it appears that the advantage to be gained from such an approach is small; away from extinction the effects of flame stretch are relatively small and the local stretch rate provides insufficient information to characterise local extinction phenomena. Even if this were not the case, the extension of the approach to LES is likely to lead to unphysical results as the distribution of subgrid "scalar" dissipation rate in space is "spotty." Since in the present case there is no experimental evidence of significant local extinction (from measurements or visual observation) a single flamelet at a low strain rate of 15 s^{-1} has been used. This laminar flamelet was constructed through computations of laminar counterflow flames using the following global mechanism [44]:

$$\begin{aligned}
&C_3H_8 + \tfrac{3}{2}O_2 \rightharpoonup 3CO + 4H_2 && \text{(i)}\\
&C_3H_8 + 3H_2O \rightharpoonup 3CO + 7H_2 && \text{(ii)}\\
&H_2 + \tfrac{1}{2}O_2 \rightleftharpoons H_2O && \text{(iii)}\\
&CO + H_2O \rightleftharpoons CO_2 + H_2 && \text{(iv)}
\end{aligned} \tag{10}$$

While there are more detailed and realistic reaction mechanisms available, the scheme (10) is found to yield acceptable results in both premixed and non-premixed laminar flames, at least under conditions far from extinction, and is deemed sufficient for present purposes.

The relationship between each thermochemical variable and the mixture fraction can be expressed as a polynomial of degree m in the conserved scalar and can be summarised as

$$\rho = \check{\rho}(\xi), \qquad T = \check{T}(\xi), \qquad Y_\alpha = \check{Y}_\alpha(\xi). \tag{11}$$

In the context of LES, the concept of "filtered density function" is introduced (see also [11]), which is essentially the probability density function of the random scalar variable ξ. This is defined as

$$\bar{P}(\Psi;\mathbf{x},t) = \int_{-\infty}^{+\infty} \mathcal{P}\big(\Psi,\check{\xi}(\mathbf{x}',t)\big)G(x-x',\Delta)\,dx', \tag{12}$$

where Ψ is the sample space of ξ, $\check{\xi}$ represents the deterministic value of the conserved scalar, and $\mathcal{P}(\Psi,\check{\xi}(\mathbf{x}',t))$ is a "fine-grained" density function [14], defined as

$$\mathcal{P}\big(\Psi,\check{\xi}(\mathbf{x},t)\big) = \delta\big[\Psi - \check{\xi}(\mathbf{x},t)\big]. \tag{13}$$

$\bar{P}(\Psi;\mathbf{x},t)$ exhibits all of the properties of a probability density function provided that the filter kernel G is positive definite [12,14]. The filtered density function is related to the density-weighted probability function $\tilde{P}(\Psi;\mathbf{x},t)$ in the following way:

$$\tilde{P}(\Psi;\mathbf{x},t) = \frac{\check{\rho}\bar{P}(\Psi;\mathbf{x},t)}{\bar{\rho}(\mathbf{x},t)}. \tag{14}$$

From this the resolved density and the resolved density-weighted thermochemical variables (temperature and species mass fractions) can be obtained:

$$\bar{\rho}(\mathbf{x},t) = \left\{\int_0^1 \check{\rho}^{-1}(\Psi)\tilde{P}(\Psi;\mathbf{x},t)\,d\Psi\right\}^{-1}, \tag{15}$$

$$\tilde{Y}_\alpha(\mathbf{x},t) = \left\{\int_0^1 \check{Y}_\alpha(\Psi)\tilde{P}(\Psi;\mathbf{x},t)\,d\Psi\right\}. \tag{16}$$

From Eqs. (15) and (16) it is easy to obtain the resolved unweighted variables, for example,

$$\bar{Y}_\alpha(\mathbf{x},t) = \bar{\rho}(\mathbf{x},t)\left\{\int_0^1 \frac{\check{Y}_\alpha(\Psi)\tilde{P}(\Psi;\mathbf{x},t)}{\check{\rho}(\Psi)}\,d\Psi\right\}. \tag{17}$$

The functional form of $\tilde{P}(\Psi;\mathbf{x},t)$ can be conveniently chosen as a β function [16], which is defined in the interval [0, 1], as is the mixture fraction itself. If the thermochemical variables are expressed as polynomial functions of ξ then the required integrals can

be obtained quasi-analytically,

$$\tilde{P}(\Psi;\mathbf{x},t)=\frac{\xi^{r-1}(1-\xi)^{s-1}}{\int_0^1 \xi^{r-1}(1-\xi)^{s-1}d\xi}, \tag{18}$$

where $r=\tilde{\xi}\{\tilde{\xi}(1-\tilde{\xi})/\tilde{\xi}''^2_{\mathrm{sgs}}-1\}$ and $s=[(1-\tilde{\xi})/\tilde{\xi}]r$.

In the present work the subgrid scalar variance is described by the gradient model

$$\tilde{\xi}''^2_{\mathrm{sgs}}=C_\xi \Delta^2\left(\frac{\partial\tilde{\xi}}{\partial x_i}\frac{\partial\tilde{\xi}}{\partial x_i}\right), \tag{19}$$

where the constant C_ξ is assigned the value 0.1 [17].

Although the effect of strain on the laminar flamelet has been ignored during this study, the resolved conserved scalar field has been employed to obtain an estimate for the subgrid "scalar" dissipation rate χ_{sgs}. Following dimensional arguments [32] χ_{sgs} can be assumed to be

$$\chi_{\mathrm{sgs}}\sim\|\tilde{S}_{ij}\|\tilde{\xi}''^2_{\mathrm{sgs}}. \tag{20}$$

With use of Eq. (20) the following expression for χ_{sgs} can be obtained:

$$\chi_{\mathrm{sgs}}=2C_s\Delta^2\|\tilde{S}_{ij}\|\frac{\partial\tilde{\xi}}{\partial x_i}\frac{\partial\tilde{\xi}}{\partial x_i}. \tag{21}$$

To complete the formulation molecular viscosities have been computed as a function of temperature and composition using the empirical relationship [45]

$$\mu(T)=\sum\frac{\alpha_i\mu_i(T)}{\sum\alpha_j\phi_{i,j}}, \tag{22}$$

where α_i is the mole fraction of species i and $\mu_i(T)$ its viscosity at temperature T. The term $\phi_{i,j}$ depends on species molar mass and viscosity.

3. Solution method

3.1. Numerical method

The in-house LES code BOFFIN (Boundary Fitted Flow Integrator) is used to solve the equations presented in the preceding sections. The code employs cartesian velocity components and a boundary conforming general curvilinear coordinate system with a colocated variable storage arrangement. It is based on a fully implicit low-Mach-number formulation and is second order accurate in space and time. For the momentum equation convection terms an energy-conserving discretisation scheme is used. All other spatial derivatives are approximated by standard second-order central differences, except for the convective terms in the filtered mixture fraction equation. A centred discretisation of these terms may result in overshoots and undershoots in the mixture fraction for cell Peclet numbers greater than 2. However, the mixture fraction must remain bounded if unphysical values of the density, temperature, and species concentrations are to be avoided. To avoid this a TVD scheme [46] is used to discretise the convection terms in the mixture fraction transport equation. Whilst it is well known that such schemes may result in the smearing out of small-scale flow features the approach has been adopted in practically all previous applications of LES to combustion (e.g., [17,39]); the use of such a method seems essential if bound violations are to be avoided. Time derivatives are approximated by a three-point backward difference scheme with variable time step. A four-step second-order time-accurate approximate factorisation method is applied to determine the pressure and ensure mass conservation in conjunction with a Rhie and Chow [47] pressure smoothing technique to prevent even–odd node uncoupling of the pressure and velocity fields. Matrix preconditioned conjugate gradient methods are used to solve the systems of algebraic equations resulting from the discretisation. A domain decomposition technique has been adopted to parallelise the code and the MPI message passing protocol implemented; the levels of parallel efficiency achieved are in excess of 90%. The code has been applied previously to the LES of a wide range of flows including plane and round jets in a cross-flow [48,49], fully developed channel [31], and pipe [50] flows, flow over a swept fence [51], and nonswirling and swirling jet flames [17,41]. Further details of the method may be found in [31,32,52].

3.2. Computational parameters

A curvilinear body-fitted grid [53] has been constructed for the present calculations featuring 1.024×10^6 cells, a section of which is represented in Fig. 3. The dilution and primary jets are also shown, indicated by zero mean mixture fraction isosurfaces. The computational mesh generator is based on a Laplace equation method and the mesh spacings and the corresponding filter width thus vary smoothly throughout the domain. The minimum and maximum cell expansion ratios are 0.6 and 1.6 though these extreme values correspond to a small number of locations—visible in Fig. 3—close to the combustor walls where the mesh is also highly skewed. Any influences that these exert have a negligible effect on the main flow field.

The inflow boundary conditions at the head of the combustor and the treatment of the fuel injector require care. Experimental and computational studies show that the prediction of the spreading angle of the fuel cone and, therefore, of the mixing pattern in the primary zone is strongly affected by the assigned flow conditions at the swirler exit. In general, the as-

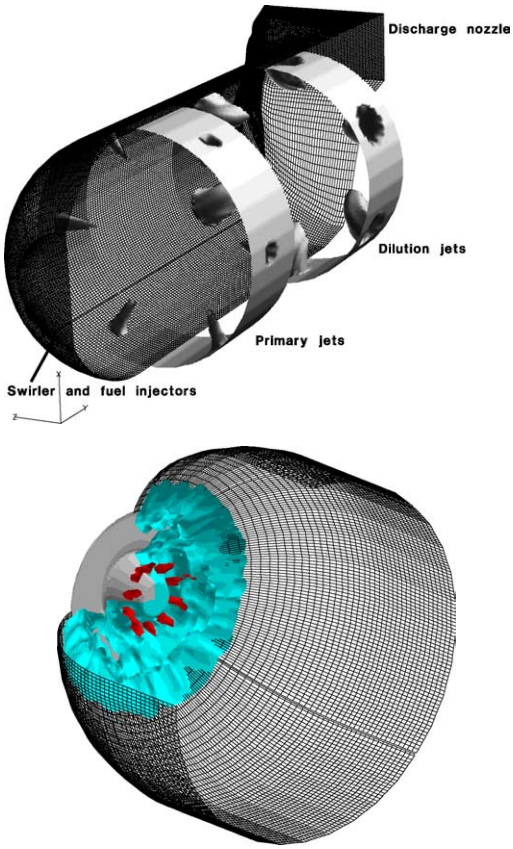

Fig. 3. Computational mesh. (Bottom) Particular of the fitted fuel injector.

sumption of a uniform velocity distribution across the swirler exit is inaccurate [29,54], and ideally account should be taken of the flow through individual blade passages and their interaction on entering the combustor. In the present case computation of the flow field upstream of and inside the swirler was not feasible and approximate boundary conditions were thus applied at the swirler exit. Two computations were carried out to investigate the influence of the swirler inlet conditions. In initial calculations (Case I) the conical section of the fuel injector was removed from the computation and replaced with a plain wall, where the fuel jets were located and uniform velocities (consistent with mass flow rates and swirl angle) were applied at the swirler exit. In a subsequent calculation (Case II) the fuel injector and fuel jets were resolved and a more detailed representation of the swirler flow was adopted [55]. An effective area at the swirler exit plane was computed and the radial velocity component was determined from the datum swirler exit area and mass flow rate. The tangential velocity was obtained taking into account the turning efficiency of the vanes and blockage effects:

$$w = \eta \frac{W_{\mathrm{sw}}}{A_{\mathrm{se}} \rho C_{\mathrm{d}} (1-b)} \tan\alpha. \tag{23}$$

In the present case the vanes are thin and a value of the blockage factor equal to 0.1 has been chosen. Based on industrial experience, values of 0.92 and 0.75 have been assigned to the turning efficiency and the discharge coefficient, C_{d}. The value of the tangential velocity obtained from Eq. (23) yields an increased swirl angle, consistent with experimental observations [54].

The proper choice of turbulent conditions at an inflow may, in certain circumstances, exert crucial influence in LES, since the influence of upstream conditions can persist downstream [56]. Several methods have been adopted in the context of LES to derive instantaneous boundary values for the velocity field, amongst which are gaussian forcing [17,32], stochastic methods [39], and inverse Fourier transform [31,57]. The merits and defects of these techniques are still a subject of debate and it has been recognised, especially in free spatially developing flows, that the behaviour of higher-order velocity correlations is strongly affected by the inner structure of the turbulence entering the computational domain [58,59]. However, application of these methods to the swirler is not practicable and so an alternate albeit crude approach is adopted. Turbulence was mimicked by adding white noise perturbations, corresponding to a turbulence intensity of 20%, to all three mean velocity components. This random signal was presumed constant over regions of size comparable to the integral length and applied at discrete time intervals corresponding to the integral time scale. The integral length scale was estimated to be half of the gap between the swirler vanes, and the time scale was evaluated using the rms of the velocity fluctuations. The procedure thus generates a turbulence field crudely correlated in space and time. As will be shown later this brought some improvements to the solution due to the resulting increased transfer of momentum in the radial direction. For the dilution jets, which are effectively plenum fed, turbulence fluctuations were neglected and uniform mean velocities were imposed at the jet inlets. Fixed mass flow rate inflow boundaries were assigned at the lateral boundaries to mimic the presence of porous walls, and convective outflow conditions were applied at the exit plane. The wall shear stresses have been calculated using approximated boundary conditions based on the logarithmic law of the wall to be satisfied in the mean [48,49,60]. The time step was chosen on the basis of accuracy considerations so that the maximum Courant number, based on the mean velocity, never exceeded 0.1 throughout the computations.

The computations corresponding to Cases I and II were carried out on a Cray T3E at the University of Manchester using 64 processing elements and on a local Beowulf cluster featuring 16 processors. With the former of these, a total wall-clock time of 26,432 h was necessary to compute approximately two residence times, corresponding to 88,944 time steps over which statistics have been accumulated; nearly twice as much time was required on the Beowulf cluster. The simulations were started from an interpolated solution obtained from a preliminary calculation on a coarse mesh, and collection of running averages [17] was commenced after the initial field had been removed from the domain ($\approx$ are residence time).

4. Results and discussion

In this section the results of the computation are discussed and compared with the measurements presented in [23,29]. Because of the limited optical assessment, only measurements of mean temperature and mean species concentrations are available for validation purposes. Velocity measurements are available only for a water model of the Tay combustor though this implies a different inflow distribution through the combustor for the obvious reason that the porous walls could not be taken into account. Therefore only a qualitative analysis of the velocity field is presented based on the computed Reynolds averaged velocity components and visual observation of the experimental configuration.

Reynolds averages and density-weighted averages (Favre averages) have been accumulated over 88,944 time steps with the former being defined as

$$\langle \phi(\mathbf{x}) \rangle \approx \frac{1}{N} \sum_{n=1}^{N} \tilde{\phi}(\mathbf{x}, t_n), \tag{24}$$

where $\phi(\mathbf{x}, t_n)$ is the nth sample of the field out of a total of N samples. A fluctuation about this mean is defined as $\phi' = \tilde{\phi} - \langle \phi \rangle$ so that the root mean square (rms) of turbulence fluctuations in $\tilde{\phi}$ is given by

$$\langle \phi' \rangle_{\mathrm{rms}} = \left(\frac{1}{N} \sum_{n=1}^{N} \phi'(\mathbf{x}, t_n)^2 \right)^{1/2}. \tag{25}$$

The difference between Favre averaged and Reynolds averaged quantities amounts to $\langle \overline{\rho' \phi'} \rangle / \langle \rho \rangle$. Results from the present computations show that this difference is negligible; however, since there is evidence [23] that probe measurements correspond to density-weighted averages it seems more appropriate to compare Favre averaged concentrations with the measured ones. From both the Reynolds and Favre averaged temperature and concentration fields, distribution at selected locations in the computational domain have been obtained as follows. For the generic variable ϕ the distributions of ensemble (Favre) averaged values, $P_s(\bar{\phi}; x)$, have been calculated according to a sampling procedure with no replacement [61], and the range of possible values was divided in 50 intervals. The standard definition of variance (or more precisely standard deviation) of a discrete distribution was then employed to compute conditional rms.

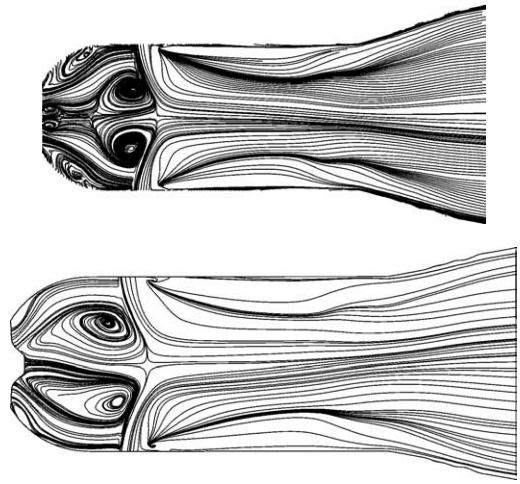

Fig. 4. Reynolds averaged flow field: streamlines in the horizontal midplane. (Top) Case I. (Bottom) Case II.

4.1. Velocity field

The isothermal flow field in the Tay combustor has been characterised by means of LDA in [29,30] and measurements have been also taken in a similar though not identical configuration [62,63]. The isothermal flow was investigated in a water model of the combustor and the presence of porous wall consequently could not be represented. The study carried out in [63] also showed that differences in the combustor's configuration (mainly in the primary zone) produced qualitatively different flow patterns.

Though a computation of the isothermal case was beyond the scope of the present study, it is still useful to consider the qualitative features displayed by the velocity field in the burning case, as shown by the ensemble averaged streamlines in the midhorizontal and vertical planes. In Figs. 4 and 5 the differences between the flow topologies predicted in Cases I and II are particularly noticeable. In both instances it can be observed that the central core vortex interacts with previous impinging primary jets, which drive the backflow in the primary region. A substantial amount of fluid is then pushed towards the wall, feeding a secondary recirculation region. These features are consistent with previous experimental observations [23,29]. However, in Case I a smaller vor-

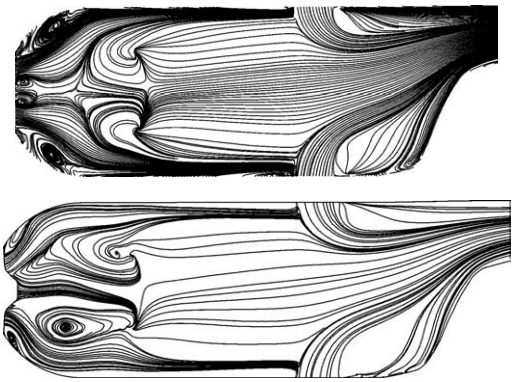

Fig. 5. Reynolds averaged flow field: streamlines in the vertical midplane. (Top) Case I. (Bottom) Case II.

tical structure is established above the fuel/air inlet which was absent from the experiments. Its formation is believed to be associated with the approximate representation of the geometry whereby the 90° conical fuelling device was replaced by a plain wall, thus approaching a bluff-body configuration. The presence of this additional toroidal vortex sensibly alters the flow configuration in the primary zone and, consequently, the predicted species and temperature distributions.

The flow field downstream of the primary jets, however, is not influenced by the changes in the geometrical description of the fuel injector and the swirler exit conditions. In particular in Fig. 6 the impingement region of the primary jets at the centre of the combustor is clearly shown. As observed in the experiments of Bicen et al. [29], this determines the strength of the backflow in the combustor head. The rotating core at the centre of the combustor is an artifact of the averaging process as the location of jet impingement point varies with time ("flapping jets"). On the other hand, the behaviour of the dilution jets is of a different nature. In fact, these maintain their individuality, as Fig. 4 shows, and the flow in the discharge region is better described as that of jets in a cross-flow. In Fig. 6 it is possible to identify a slight asymmetry in the penetration of the jets and this may be associated with the circular to rectangular discharge nozzle immediately downstream of the dilution ports. Details of mixing in the proximity of the nozzle are discussed further in the following sections.

Consideration of the flow visualisation results of Bicen et al. [29] and Bicen and Palma [30] suggests that the computations correctly reproduce the main features of the flow field. Also, and perhaps as expected, the predicted flow field obtained from a preliminary computation on a coarse mesh (125,000 cells) did not differ from that produced with the refined mesh. This suggests that even on the coarse mesh most of the turbulent energy has been accounted

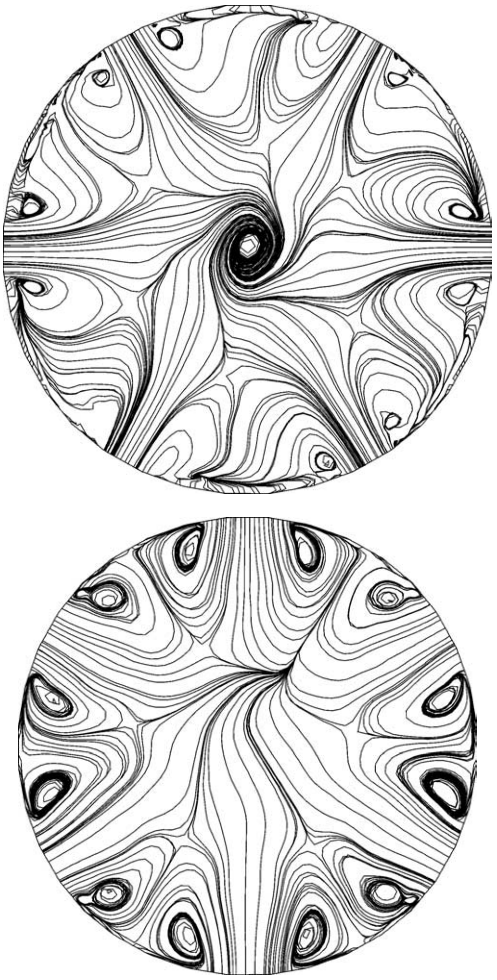

Fig. 6. Reynolds averaged flow field, Case I. (Top) Primary port cross section. (Bottom) Dilution port cross section.

for and that a different choice of the SGS model would not sensibly affect the predictions. These considerations are clearly valid only as long as near-wall dynamics (e.g., heat transfer at the walls) are not of interest.

4.2. Species concentrations and temperature

The flow pattern described in the previous section can be immediately recognised in the time evolution of the temperature field (Case I) as shown in the sequence of six images presented in Fig. 7. The hot regions in the primary zone and along the sides of the combustor, between the primary and dilution jets, reproduce faithfully the experimental findings. In fact, no combustion occurs beyond the dilution ports, and the flame is clearly stabilised in the recirculation region in the head of the combustor and in the wake of the primary jets. No reaction processes reach the liner

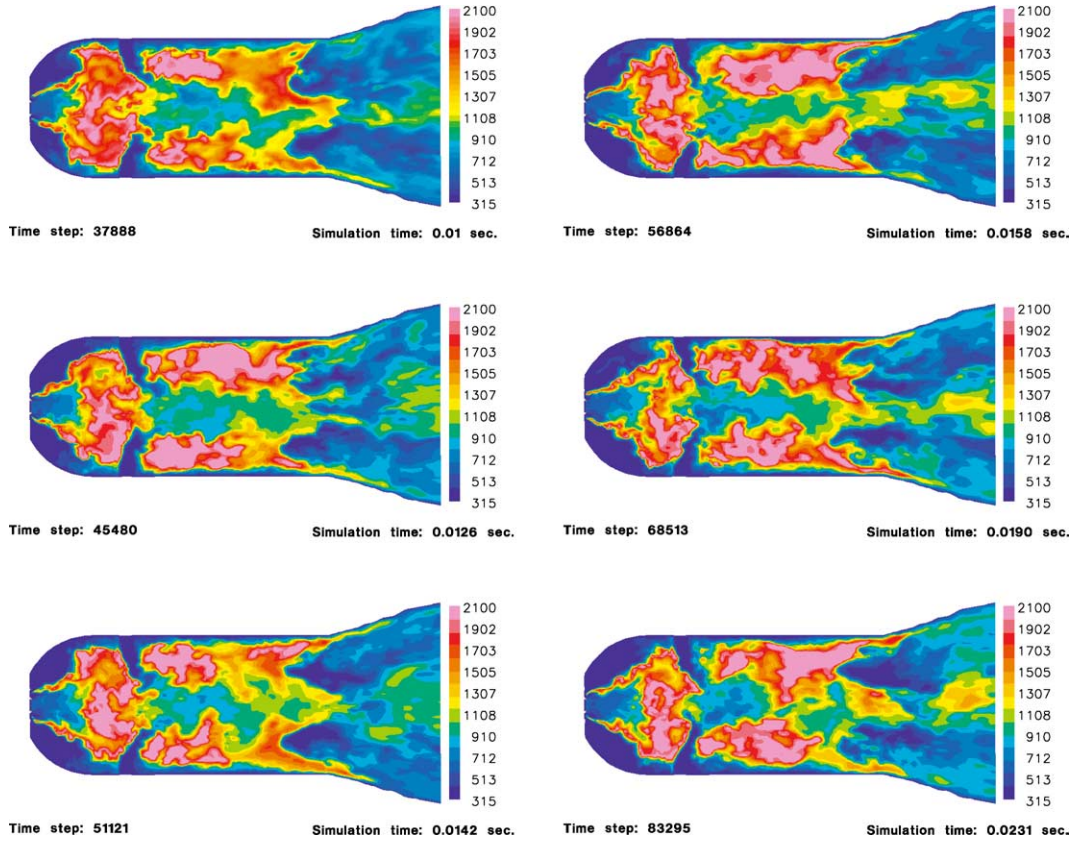

Fig. 7. Instantaneous temperature field: horizontal midplane.

walls, which is isolated by the cooling film of air passing through the *Transply* layer that is clearly visible in all the pictures. Also evident is the occasional release of packets of hot gases that manage to escape the primary region and reach the discharge nozzle. Flow visualisation studies suggest that these events are related to the unsteadiness around the primary jets impingement point and that they exert important influence on the temperature distribution at the exit plane. This can be qualitatively characterised by the pattern (OTDF) and profile (RTDF) factors [64] expressed as

$$\mathrm{OTDF} = \frac{(T_{\mathrm{max}} - T_{\mathrm{mean}})}{(T_{\mathrm{mean}} - T_{\mathrm{inlet}})}, \tag{26}$$

$$\mathrm{RTDF} = \frac{(T_{\mathrm{mean,radius}} - T_{\mathrm{mean}})}{(T_{\mathrm{mean}} - T_{\mathrm{inlet}})}. \tag{27}$$

The computed pattern factor for Case I is 0.53, a value somewhat larger than the measured value of 0.46. The computed RTDF was 0.15 at 65% of turbine blade height compared with 0.2 at 75% observed in [23]. The results obtained in Cases I and II were practically identical.

The discrepancy between the computed and measured OTDF and RTDF can be explained by an underprediction of the penetration of the dilution jets in the lower part of the barrel. It appears that the large acceleration experienced by the flow in the longitudinal direction excessively inhibits the mixing of the hot gases coming from the primary and intermediate regions with the cold dilution jets. It is unlikely that this can be attributed to the hydrodynamic closure and it seems more likely that this is a result of the oversimplified "top hat" specification of the velocity field at the dilution jet inlet. It is probable that the pressure distribution within the combustor distorts the incoming velocity profile at the dilution ports. A similar effect has been observed in simulations of round jets in a cross-flow [31,49].

Although the instantaneous temperature patterns are qualitatively similar in both Cases I and II, comparisons of the density-weighted temperature contours in the horizontal midplane (Fig. 8) clearly illustrate the influence of the swirler inflow conditions and the fuel injector geometry on the predicted large-scale mixing in the primary zone. In Case I, as can be inferred from the measured temperature and composition profiles in the primary zone, insufficient mixing of the fuel and swirler air is predicted so that the fuel cone closes on itself and the stoichiometric mixture

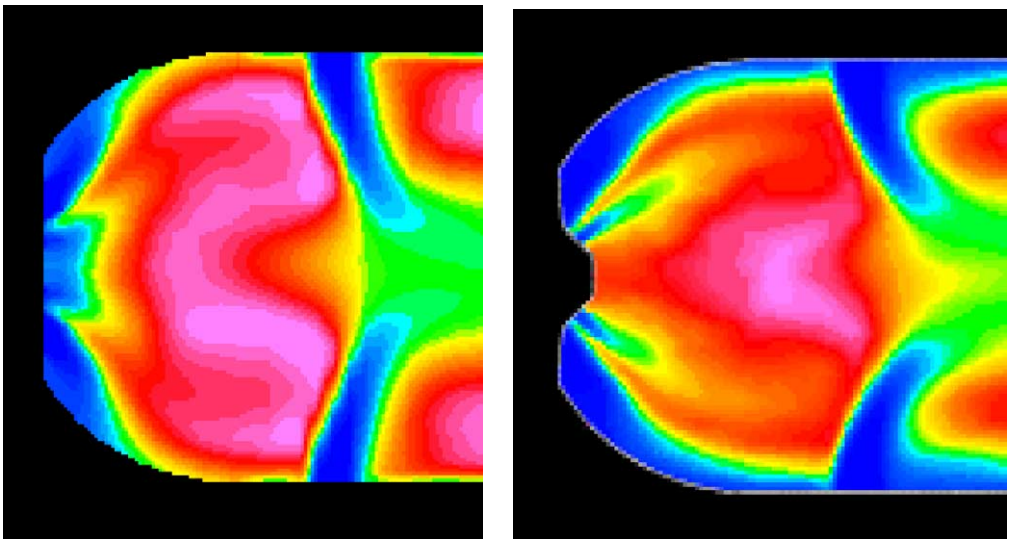

Fig. 8. Density-weighted temperature in the primary zone: horizontal midplane. (Left) Case I. (Right) Case II. Colour scale: five levels between pink = 2200 K and blue = 315 K.

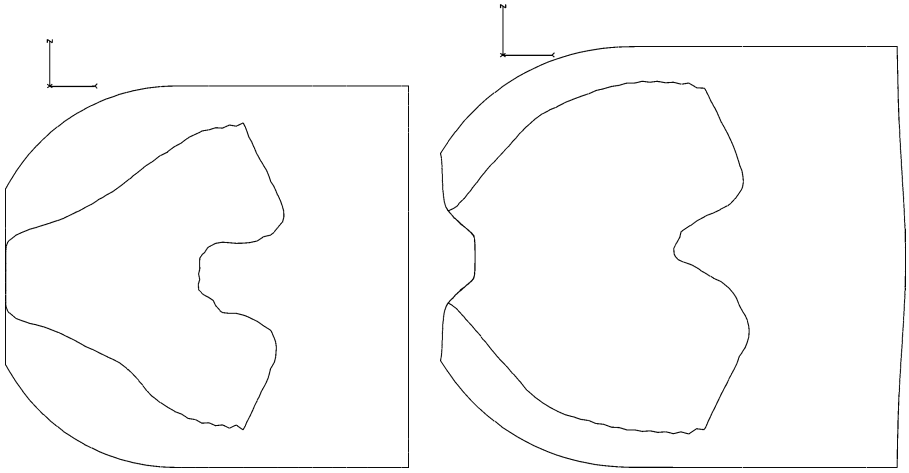

Fig. 9. Locus of stoichiometric density-weighted mixture fraction in the primary zone: horizontal midplane. (Left) Case I. (Right) Case II.

fraction contour lies too close to the fuel injector. The situation changes considerably with the more realistic inflow and geometry of Case II where much more plausible results are obtained, as is illustrated in Fig. 9 where the loci of the stoichiometric mixture fraction are shown.

The influence of the swirler inlet conditions is also evident in the mean temperature profile at $z = 20$ mm shown in Fig. 10; and while the results of Case II are appreciably closer to the measured profile some discrepancies remain. It, however, appears that the profiles downstream of the primary zone are largely unaffected by the conditions at the head of the combustor. The profiles at $z = 50$, 80, and 130 mm are in reasonably good agreement with measured values and show that the temperatures in the proximity of the wall are kept below 700 K by the presence of the resolved cooling film.

To investigate further the effect of inlet conditions a computation was completed with *no* turbulence fluctuations imposed on the incoming swirler flow. An example of the effects can be seen in Fig. 11, where profiles of the Reynolds averaged mixture fraction (though no data are available for comparisons) are shown in the primary zone at $z = 20$ mm and downstream of the primary dilution jets at $z = 50$ mm. It appears the that the choice of random noise, albeit crudely correlated, is not adequate and that a more

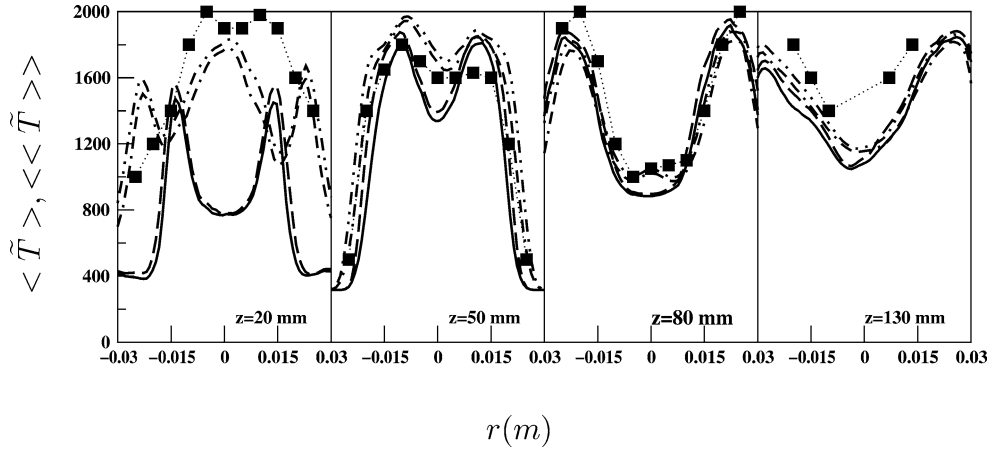

Fig. 10. Profiles of Reynolds and density-weighted averaged temperature in the horizontal midplane. Solid line: Case I, density-weighted average; dashed line: Case I, Reynolds average; double-dashed dotted line: Case II, density-weighted average; double-dotted dashed line: Case II, Reynolds average. ■: [23].

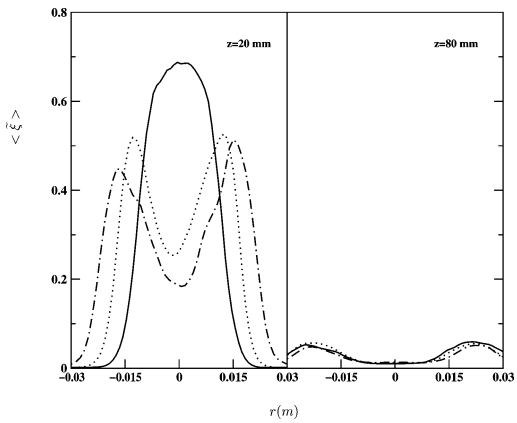

Fig. 11. Profiles of Reynolds averaged mixture fraction in the horizontal midplane. Solid line: Case I; dot-dashed line: Case II; dotted line: same as Case II, with zero turbulence intensity.

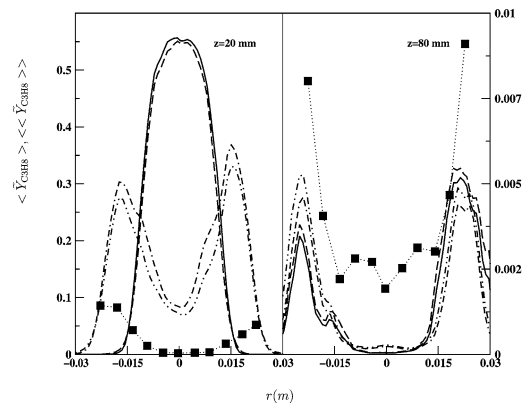

Fig. 12. Profiles of Reynolds and density-weighted averaged C_3H_8 mole fraction in the horizontal midplane. Solid line: Case I, density-weighted average; dashed line: Case I, Reynolds average; double-dashed dotted line: Case II, density-weighted average; double-dotted dashed line: Case II, Reynolds average. ■: [23].

realistic representation at the swirler exit of turbulent fluctuations and probably the mean profile also is needed.

A trend consistent with that displayed by the mixture fraction and temperature can be observed in the concentrations of C_3H_8, CO_2, and O_2 (Figs. 12–15). Although predictions generally improve at $z = 80$ mm, it is clear that CO levels are not well represented by the current flamelet assumption. On the other hand, the temperature and the other species seem to be less sensitive to this and are more strongly influenced by transport effects. It seems likely that the poor prediction of the mixture fraction field magnifies the shortcoming embedded in the flamelet formalism, and that the use of a more detailed reaction mechanism would have little effect on the prediction of CO concentrations if used in the context of a conserved scalar approach. This is likely to be the case even if strain effects were to be included in the formalism as the values of "scalar" dissipation are low throughout much of the combustor.

If flame stretch effects were to be included then this would require the joint pdf of mixture fraction and the "scalar" dissipation rate. Such an approach was adopted in [39], where an unsteady flamelet formulation was incorporated into LES of a turbulent diffusion jet flame (Sandia Flame D), with the "scalar" dissipation rate being used to provide a measure of flame stretch effects. Although such a model has not been applied in this work, an estimate of $\langle\chi\rangle = \langle\chi_{\text{sgs}}\rangle + \langle\tilde{\chi}\rangle$ has been obtained according to the model

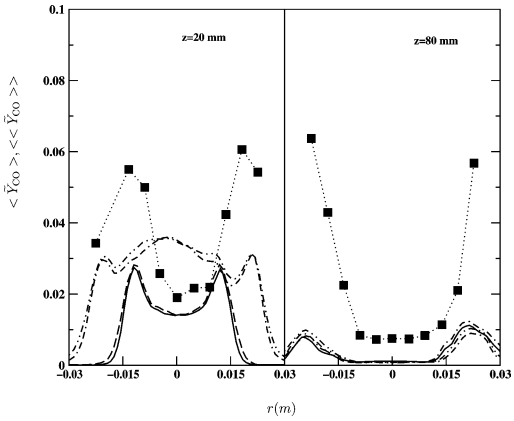

Fig. 13. Profiles of Reynolds and density-weighted averaged CO mole fraction in the horizontal midplane. Solid line: Case I, density-weighted average; dashed line: Case I, Reynolds average; double-dashed dotted line: Case II, density-weighted average; double-dotted dashed line: Case II, Reynolds average. ■: [23].

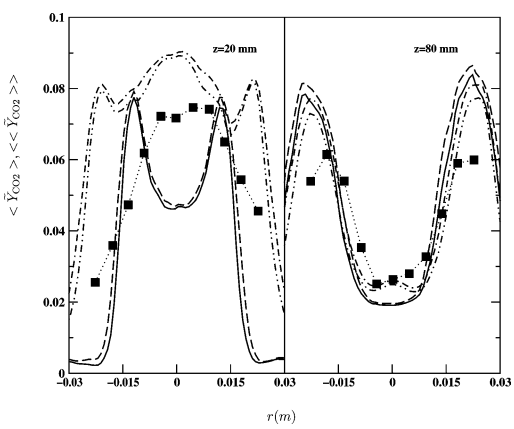

Fig. 14. Profiles of Reynolds and density-weighted averaged CO_2 mole fraction in the horizontal midplane. Solid line: Case I, density-weighted average; dashed line: Case I, Reynolds average; double-dashed dotted line: Case II, density-weighted average; double-dotted dashed line: Case II, Reynolds average. ■: [23].

presented in the previous sections. In Figs. 16 and 17 contours and scatterplots of $\langle\chi\rangle$ in mixture fraction space are shown over cross sections of the combustor at relevant locations along the axis. It is interesting to note that very low values of "scalar" dissipation rate prevail throughout the computational domain, becoming negligible as the secondary region, downstream of the primary dilution jets, is approached and that values of the mixture fraction span an even narrower range around the lean limit. The contours of $\langle\chi\rangle$ also serve as indicators of the relevant structures present in the flow field and of the large differences in the mixing pattern in the primary zone arising in Cases I

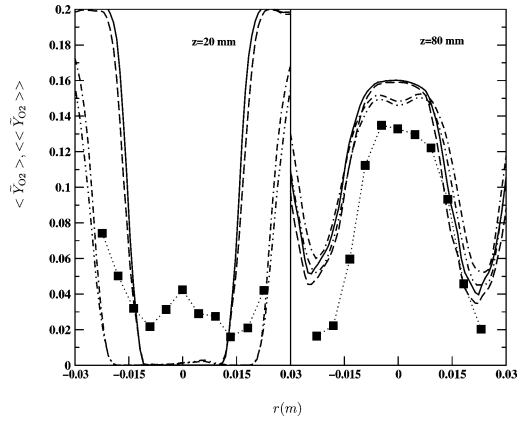

Fig. 15. Profiles of Reynolds and density-weighted averaged O_2 mole fraction in the horizontal midplane. Solid line: Case I, density-weighted average; dashed line: Case I, Reynolds average; double-dashed dotted line: Case II, density-weighted average; double-dotted dashed line: Case II, Reynolds average. ■: [23].

and II. The low levels of "scalar" dissipation evident from the computations are consistent with the experimental observation that negligible local extinction phenomena occur in the combustor.

5. Conclusions

Results of the large eddy simulation of a model can-type gas turbine combustor have been presented. The chosen geometry retained all of the features of can-type combustion chambers commonly encountered in commercial aeroengines, and offered an excellent test case to prove the potential of LES to accurately predict complex, three-dimensional unsteady flows. While no velocity measurements are available for comparison purposes the computed flow pattern is in reasonable agreement with that observed experimentally. However, mixing rates in the primary zone are shown to be strongly dependent on the inlet conditions at the swirler exit. When a uniform profile with no turbulence fluctuations was imposed with the fuel injector being represented by an annular slot, primary zone mixing rates were substantially underrepresented. As a consequence large discrepancies between the calculated and measured profiles of temperature and C_3H_8 are evident in the primary zone. The situation was much improved when more realistic inlet conditions were applied though some disagreement, particularly with respect to the levels of C_3H_8, remains. It is likely that the estimated mean profiles and simple method of estimating turbulent fluctuations are not entirely satisfactory and what is probably required is for the solution domain to be extended

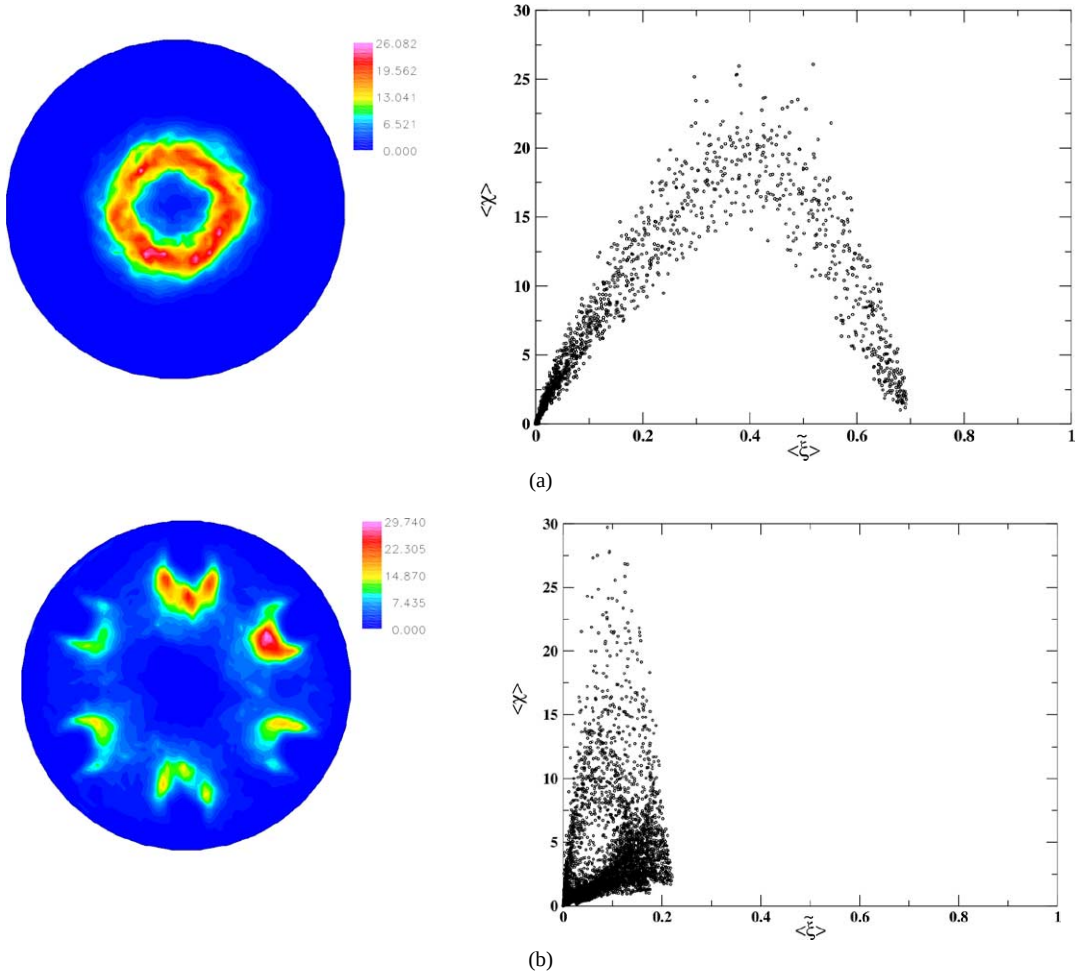

Fig. 16. Contours and scatterplots of total ($\langle\chi_{\text{sgs}}\rangle + \langle\tilde{\chi}\rangle$) scalar dissipation rate: Case I. (a) $z = 20$ mm, (b) $z = 50$ mm.

back through the swirler and upstream of the combustor.

The extent of the influence of the inlet conditions in the head of the combustor is limited by the primary dilution jets, which impinge on the combustor centreline. Consequently it is likely that the conditions at the primary jet, swirler, and fuel inflows play little major role in the development of the turbulence structure in the region downstream of the primary jets. Consistent with this, but with the exception of CO, the agreement between measured and computed mean temperature and composition profiles is reasonable downstream of the primary zone. Also, the magnitude of the SGS stresses seems to suggest that very little sensitivity to the SGS closure is displayed by the resolved scales. This result is encouraging and suggests that the spatial resolution is adequate. Evaluation of the computational mesh and filter width variation also suggests that commutation- and cell anisotropy-related errors are small and have little effect on the predicted velocity and scalar fields.

Only CO concentrations seem to be particularly affected by the description of the combustion process described via the flamelet model, whilst temperature and other species appear more sensitive to the large-scale transport processes. The measured levels of CO are substantially higher than those predicted by the current flamelet model in the region downstream of the primary jets. While the effect of flame stretch is to increase CO levels these effects are almost certainly negligible in this downstream region. It thus is appears unlikely that any steady flamelet/conserved scalar model is capable of reproducing these observed CO levels. The measured levels of CO are more likely to arise because of the reaction quenching effect of the cold dilution jets. If this is the case then the "fast" reaction assumption inherent in the flamelet/conserved scalar model must be relaxed.

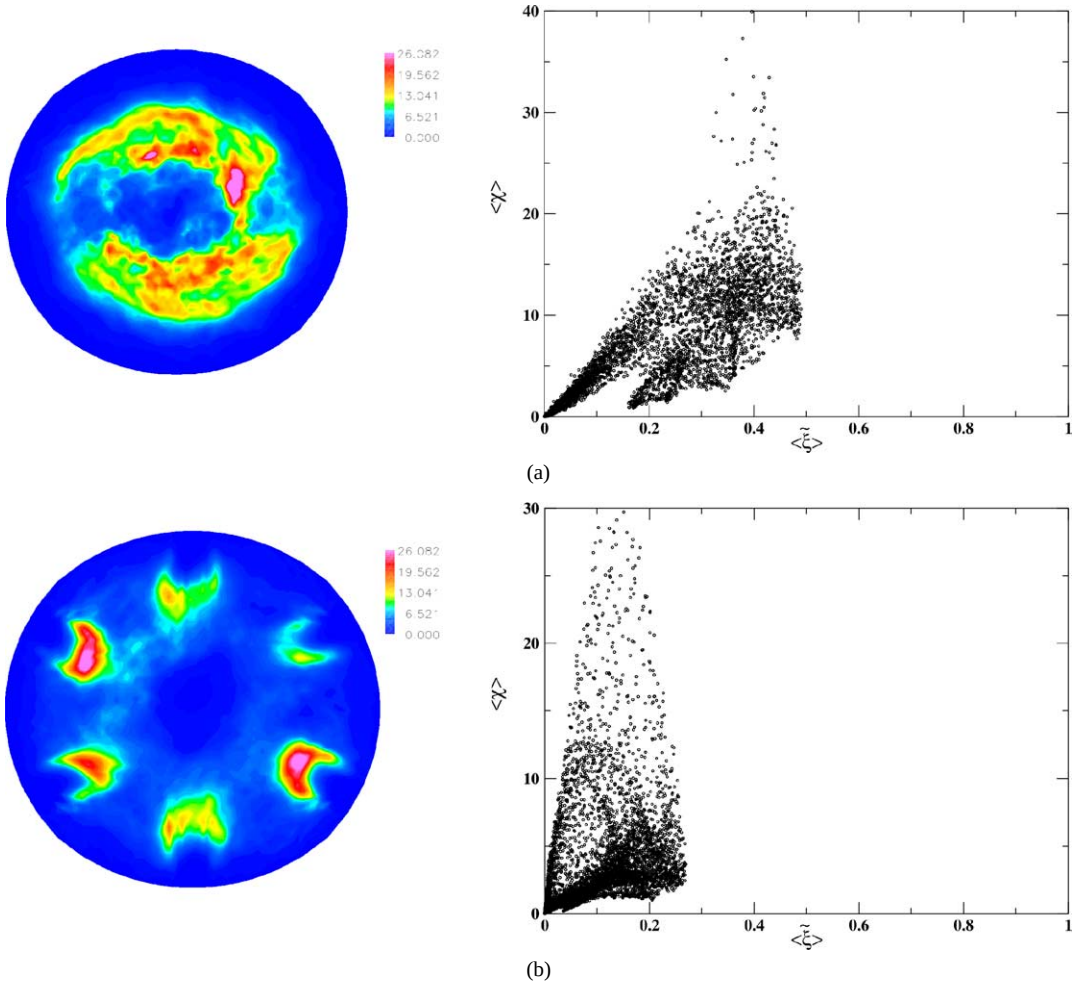

Fig. 17. Contours and scatterplots of total $(\langle\chi_{\mathrm{sgs}}\rangle + \langle\tilde{\chi}\rangle)$ scalar dissipation rate: Case II. (a) $z = 20$ mm, (d) $z = 50$ mm.

It could be argued that a single laminar flamelet (corresponding to a very low strain rate) is not sufficient to describe combustion throughout the entire domain. However, the distribution of the estimated scalar dissipation rate seems to suggest that this parameter would play a very limited role and the computed levels are consistent with the experimental observation that no significant local extinction occurs in the present case. Of course, in general the possibility of local extinction/reignition phenomena should be taken into account. However, the possibility of utilising a "resolved" scalar dissipation rate to describe the effects of stretch on laminar flamelets is somewhat debatable, since this is likely to yield an unphysical intermittent behaviour of the flame, decoupled from transport and reaction processes responsible for extinction/reignition. The magnitude of the estimated SGS stresses and scalar fluxes also seems to suggest that the majority of velocity and scalar fluctuations have been explicitly accounted for, and that the closure adopted for the SGS contributions is likely to exert but a little influence on the predictions.

Acknowledgment

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